Introduction to Big Data

Big Data Application: COVID-19

- Important tool for healthcare industry experts and decision makers
- Thermal scanners were used to identify infected individuals
- Close contact detector applications developed to notify health authorities
- Travel notifications provided by telecom companies that list places someone had visited during a 14-day period
- Dashboards with near-real time statistics on the spread and status of the disease
- Amazon Web Services, Google Cloud, and others offered researchers free access to open datasets and analytics
- Biomedical research developed advanced artificial intelligence (AI) applications that utilized Big Data sets to develop a vaccine

Chapter Objectives:

- Explain the concept of Big Data and the factors that led to the era of Big Data
- Compare and contrast the Big Data and traditional data paradigms
- Provide examples of organizations or sectors that are implementing Big Data projects
- Discuss the challenges that the organizations face and offer practical recommendations for implementing Big Data projects

Big Data:

- Exhibits all characteristics of traditional data and more
- Includes both structured and unstructured data
- Big Data Analytics is different from traditional data analytics
- Captures the large amount of data generated in:
  - *Healthcare*
  - *Internet of Things*
  - *Financial companies*
  - *Supply chain systems*
  - *Scientific sensors*
  - *Social media*
  - *Product digitization*

Big Data: A Historical Perspective

- Several events have fueled the growth of Big Data:
  - The invention of the tabulating calculator in the nineteenth century
  - The production of computers in the 1960s and 1970s
  - The advent of the Internet and the World Wide Web in the 1980s
  - The growth of electronic commerce at the beginning of the twenty-first century
  - Recent developments with the Internet of Things (IoT)
  - Other technologies in the last two decades

Table 1-1: Big Data Evolution in Historical Context

Year	Description
In 1881	The Hollerith Tabulating punch card machine was invented. The device was able to compile and tabulate the large datasets of the 1880 U.S. census within a year, a job that was estimated to take ten years because of the manual nature of the task.
In 1965	The U.S. government built the first data center and stored over 742 million tax returns and 175 million sets of fingerprints. The project was the beginning of the electronic data storage era.
In 1989	The World Wide Web (www) was invented to facilitate the sharing of information via a hypertext system.
In 2001	Industry analyst Doug Laney produced a research paper outlining the now widely accepted "3 Vs" of Big Data.
In 2003 and 2004	Two seminal papers by Google researchers were published on distributed file systems and MapReduce.
In 2005	"Roger Mougalas from O'Reilly Media coined the term Big Data."
In 2005	Yahoo! created Hadoop "on top of Google's MapReduce. Its goal was to index the entire World Wide Web."
2005-2010	Social networks started appearing. Web 2.0 took flight. Innovative startups started working on Big Data projects.

Table 1 (cont.)

Year	Description
In 2009	The Indian government created "the largest biometric database in the world," which stores iris scans, fingerprints, and photographs of 1.2 billion inhabitants.
In 2010	Eric Schmidt stated, "there were 5 exabytes of information created by the entire world between the dawn of civilization and 2003. Now that same amount is created every two days."
In 2011	McKinsey reported that in 2018, the United States would face a shortage of 140,000–190,000 data scientists as well as 1.5 million data managers.
Today	"Poor data across businesses and the government costs the U.S. economy $3.1 trillion a year," and it "costs the U.S. almost 20% of the nation's entire GDP."
Today	Every minute: "Netflix users stream 694,444 hours of video Instagram users post 277,777 stories YouTube users watch 4,500,000 videos Twitter users send 511,200 tweets Skype users make 231,840 calls Airbnb books 1,389 reservations Uber users take 9,772 rides Tinder users swipe 1,400,000 times Google conducts 4,497,420 searches"

Big Data Paradigm

- Started after 2010
- Amount of data in the world reached its first **zettabyte** (ZB)
- In 2012, the amount of information in the world reached 2 ZB
- Currently, the world has already stored more than **17 ZB** of data
- By 2030, the world is expected to reach **1000 ZB** or one **yottabyte** (YB)

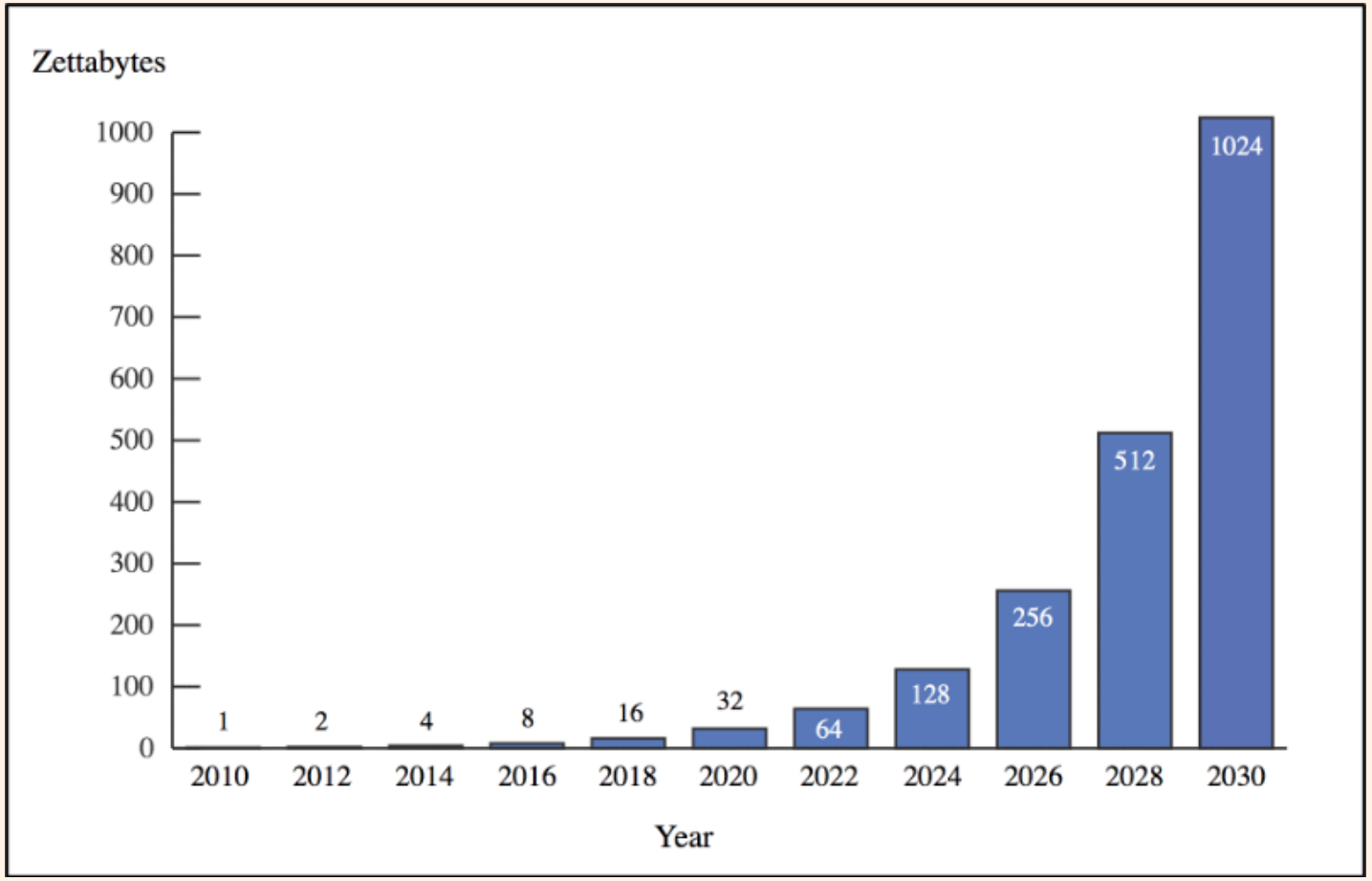

Figure 1-1 Estimated Data Growth

Big Data Definition

- Structured:
  - In-house operational **databases**
  - Automatically captured

- Unstructured:
  - From social networks
  - Web server logs
  - Banking transactions
  - The content of web pages
  - Financial market data

Big Data Definition (cont.)

- Three attributes distinguish Big Data from traditional data:
  - Volume
  - Velocity
  - Variety

Volume

- **Larger** than the volume processed by conventional relational databases
- **Data warehouses** store **structured data** in *unnormalized tables* for further data processing and analytics
- Gathered unstructured data stored in:
  - Document,
  - Column-based, or
  - Graph-based databases
- **Structured & unstructured** data => **high volume** of data

Velocity

- The **rate** at which data flow into an organization
- Online sales, mobile computing, smartphones, and social networks have increased the information flow
- Organizations **analyze and produce** operational **business intelligence** quickly and make recommendations
- Velocity of system output is **critical**
  - Example: a loan officer could compare the information in a loan application against a data mining model and make a decision about the loan

Variety

- **Refers to the mix of different data sources** in different formats
- The **format can be different**, even in a **structured** data source
- Different users may withhold information
- There may be *errors, redundancy, and inconsistency* when humans are involved

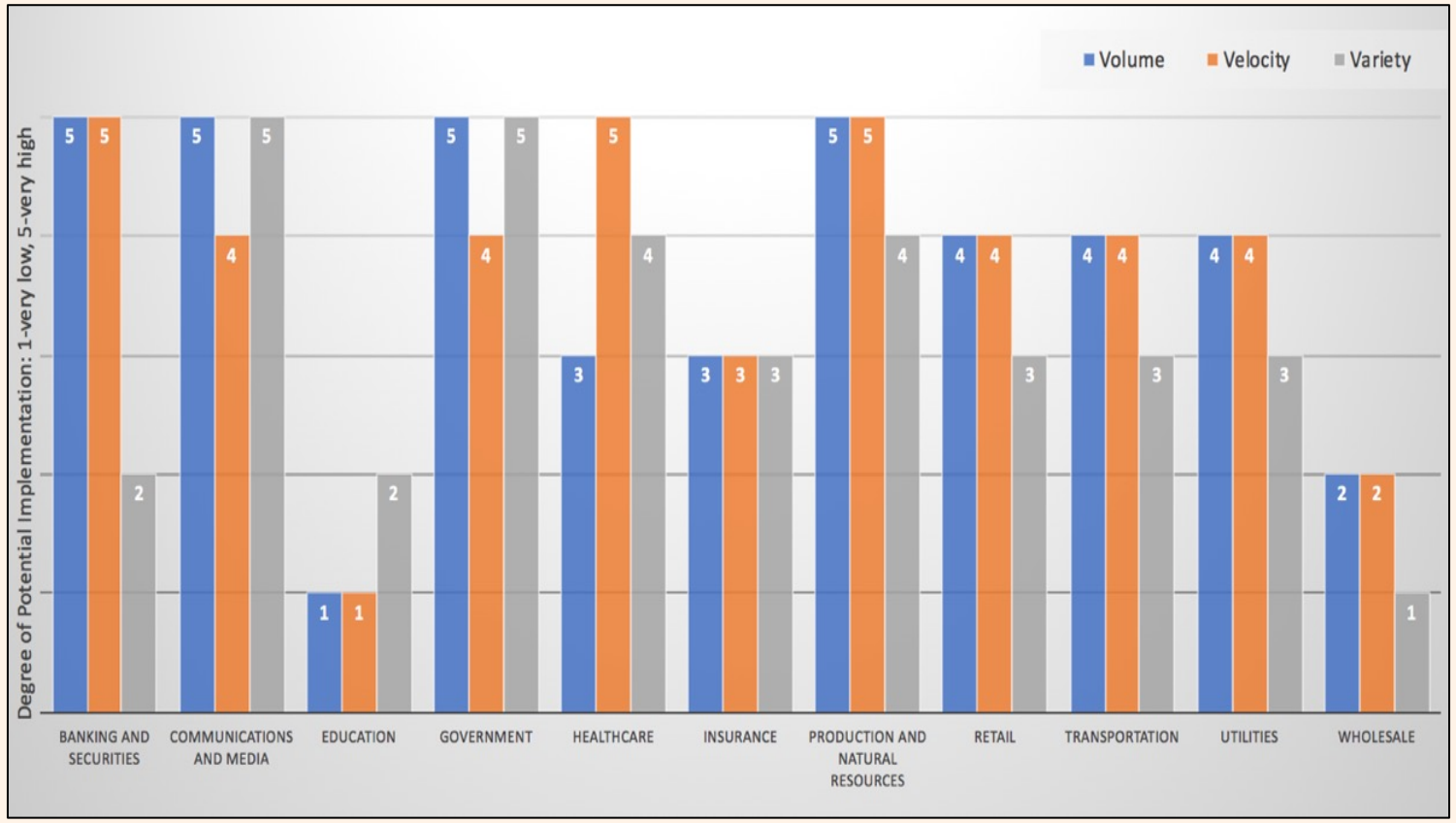

Figure 1-2: Volume, Velocity, and Variety of Data by Industry

Figure 1-2 (cont.)

- **Volume** is predominant dimension in:
  - Banking
  - Communications
  - Government
  - Production
  - Natural resources (including supply chain systems)

Figure 1-2 (cont.)

- **Velocity** is important dimension in:
  - Banking
  - Healthcare
  - Production
  - Supply chain systems

Figure 1-2 (cont.)

- **Variety** is dominant in:
  - Communications
  - Media
  - Government

Big Data Definition (cont.)

- National Institute of Standards and Technology (NIST)
  - Offers an alternative, architectural explanation of Big Data:

"Big Data is where the data volume, acquisition velocity, or data representation limits the ability to perform effective analysis using traditional relational approaches or requires the use of significant horizontal scaling for efficient processing."

Big Data Analytics in Action

- Big Data applications in the organization:
  - Structured data by:
    - **Customer relationship management** (CRM) systems
    - Points of sale
    - Human resource systems
    - Inventory systems
  - Unstructured data:
    - Internal
    - External

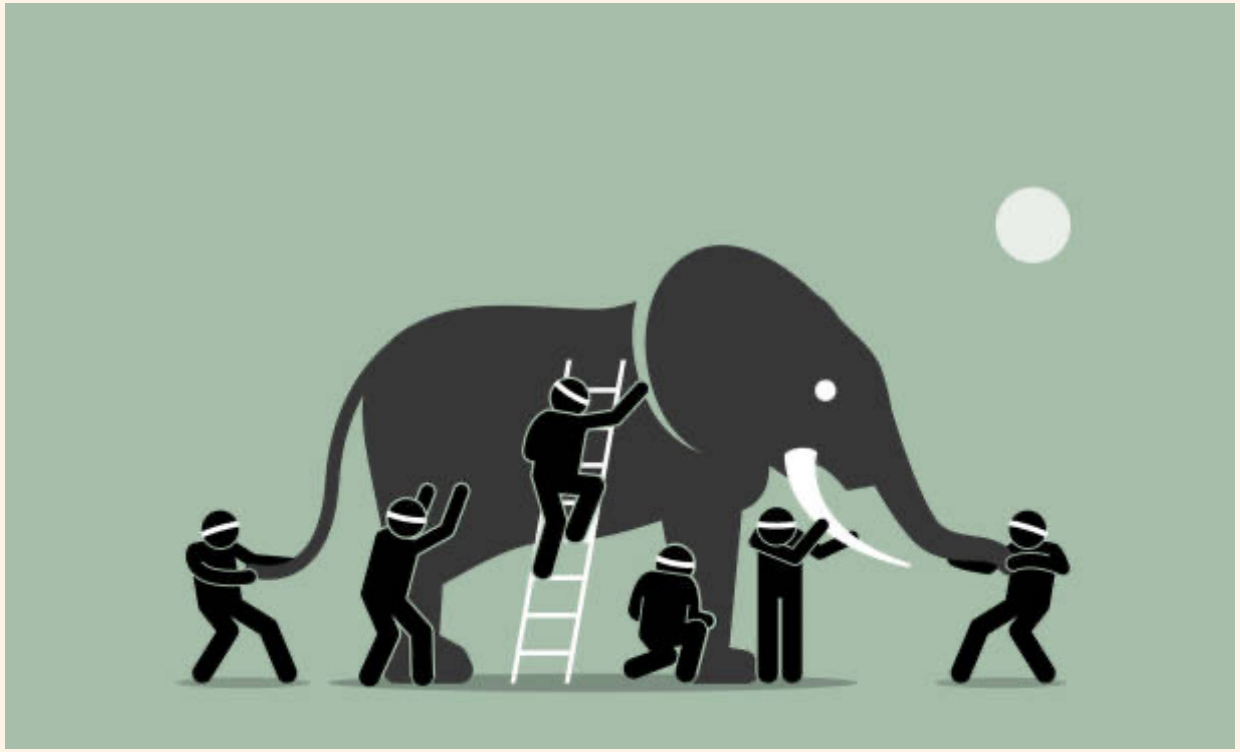

Figure 1-3 Understanding the Big Picture

Using Big Data Analytics

- Generate business insights
- Improve customer experiences
- Lower the costs of products and services

Finance

- **25%** of Big Data are generated by the financial industry
  - *New York Stock Exchange* => **1 terabyte** of information every day
  - Commercial banks collect a large amount of data, generated by:
    - Credit cards
    - Deposits
    - Loans
    - Mortgages
- Big Data techniques:
  - Fraud detection
  - Automatic trading algorithms
  - Sentiment analysis to improve client retention

Telecommunications

- **Mobile communications**—a significant source of Big Data:
  - About **45%** of carriers with active Big Data initiatives **use location services**
  - **Location services** in the mobile industry => **1 terabyte** of data every day
  - **T-Mobile** increases customer retention by integrating Big Data with its transaction systems and social media

Retail

- In 2010, **Walmart** handled **more than 1 million** customer transactions
  - Generated over **2.5 petabytes** of data every hour
- With Big Data, McKinsey & Co. estimates:
  - The retailer's operating margin **increased by more than 60%**
- Using Big Data combined with data from the Internet and social media, retailers:
  - Predict trends
  - Suggest products or services to customers
  - Optimize pricing and promotions

Utilities

- Generate a **high volume of structured data**
  - Electricity customers in the United States have smart meters
- Utility companies apply algorithms for:
  - Anomaly detection
  - Collaborative filtering
  - Energy control

Automakers

- Two significant sources of Big Data:
  - Auto-enthusiast websites
  - Forums and vehicle sensors
- Ford uses **text-mining algorithms** to analyze **more than 10,000** mentions
- Today's vehicles contain more than **10,000 sensors, which** generate data about:
  - Fuel efficiency
  - Tire wear
  - Oil pressure
  - Oil viscosity
  - Engine or water temperature

Smart Cities

- Los Angeles implemented **dynamic pricing systems and adjusted parking rates** to influence parking behavior:
  - **15%** decrease in parking occupancy in congested areas
  - **10%** increase in less desirable areas
- In New York City, trees destroyed by hurricanes were removed faster using information generated by Big Data
- In Portland, Oregon => optimal timing of traffic lights
  - Eliminated more than **157,000 metric tons of CO_2 emissions in 6 years**
- In Oslo, Norway => reduction in street lighting energy by **62%**

Package delivery

- Utilizes Big Data to improve:
  - Route delivery
  - Fuel efficiency
  - Driver safety
- Example: UPS
  - Uploads sensor data from more than 46,000 vehicles every night
  - Saved 85 million miles
  - Reduced fuel consumption by 8.4 million gallons

Sports Analytics

- Big Data for dynamic ticket pricing
  - Example, the San Francisco Giants:
    - Predicted attendance
    - Priced tickets
    - To keep the attendance up for less popular games
- Decisions regarding:
    - Scheduling
    - Player contracts
    - Ticket sales
    - Television and digital media deals
    - Merchandise sales

Healthcare

- Electronic patient medical records
- Data analysts aggregate information about healthcare outcomes
- Reveal patterns => more effective treatments and detection of outbreaks
  - **20% decrease** in inpatient mortality
  - **Over $300 billion increase** in value to the U.S. healthcare industry

Online Dating

- About **40%** of the single population
  - **Match.com**
    - Uses Big Data to improve the algorithm for matching people on dates consistently
    - Has each new user answer about 400 questions
    - Uses these answers to compare each user with about 20 million other profiles

Video Games

- **Important**: understanding the top players
  - **Top 1%** of players count typically for **25–50% of industry's revenue**
- **Zynga**, the San Francisco-based game maker of FarmVille
  - Stores **25 terabytes** of data per day
  - Uses these data to provide better:
    - Customer service
    - Quality assurance
    - Product design—deciding what features show up in the next generation of games

Implementing Big Data Projects

- Because of Big Data:
  - *A 2019 survey* of 65 leading companies from the **Fortune 1000**:
    - **91.6%** of the organizations were **increasing the pace of adoption**
    - **55%** of firms were investing **greater than $50 million**
    - Adoption of Big Data and artificial intelligence projects remained a significant challenge

Starting Big Data Projects

- First set of data storage platforms:
  - Capture, process, and store data
  - Belong to the supply side of the data
    - Examples: RDBMSs, CRM systems, inventory management database (IMD) systems, supply chain management (SCM) systems
  - Data analytics platforms are embedded in data storage platforms
    - Example: a data manipulation component such as SQL is an essential part of a CRM database management system (DBMS)

Starting Big Data Projects (cont.)

- Second set of platforms (data analytics platforms):
  - Query and analyze data
  - Belong to the demand side of the data
    - Examples:
      - Data warehouses
      - Spreadsheet platforms
      - SQL systems
      - Data analytics software
        - Such as SPSS, SAS, or Tableau

Starting Big Data Projects (cont.)

- The *quest* for Big Data projects starts:
  - The current data analytics platforms are not able to process the
    - Volume
    - Variety
  - Of incoming data generated by the data storage platforms
  - The supply is more than the demand
- *Solution:*
  - Add more data processing capacity

Selecting a Big Data Project

- Organizations must:
  - Consider an overall Big Data strategy
  - Evaluate the global scope and efforts of Big Data projects
  - Implement that strategy one project at a time
- Business activities generate a large amount of data
  - Valuable resource for generating business insights and remaining competitive
- Needs:
  - Increase the demand side of data
  - Consider implementing Big Data projects

Table 1-2: Cost-Benefit Approach to Selecting a Big Data Project

	Project Name	Estimated Cost: (Hardware + Software + Skills)	Likely Annual Benefit	Ratio: Benefits/Cost	Priority
1	Analyzing tracking devices to optimize playtime	$200,000	$50,000	0.25	3rd
2	Social media content analysis	$150,000	$30,000	0.20	4th
3	Using digital sensors for better product design	$300,000	$80,000	0.27	2nd
4	Health vs. safety: combining business transactions records with health tracking	$80,000	$70,000	0.88	1st

Selecting a Big Data Project (cont.)

- The project priorities are defined:
  - By the ratio of the potential benefits to the estimated costs
  - The higher the ratio, the higher the priority of the Big Data project
- Other quantitative models for selecting a Big Data project include:
  - The project scoring model
  - Payback period
  - Net present value
  - Discounted cash flow
  - Internal rate of return
  - Opportunity cost

Managerial Considerations

- C-suite of the organization must support Big Data Project
  - *Project aligns better with the overall organizational strategy*
  - Deal with organizational changes associated with the new Big Data culture
- The information technology department manages data analysts and data analytics platforms
- Data analytics projects are initiated and used by departmental functions or business units

Sourcing of Big Data Projects

- Outsourcing Big Data initiative:
  - Growing at a compound annual rate of 29.7%
  - To achieve a $40.6 billion value by 2023 and a $229.4 billion value by 2025

Sourcing of Big Data Projects (cont.)

- Insourcing:
  - Considered a better alternative because of the nature of data analytics
  - Data is unique
  - Organizations use Big Data projects to create products and services that are unique, rare, hard to imitate, and hard to replace
  - Outsourcing can provide a general formulation and solution to an organization
  - On the other hand, insourcing:
    - More expensive at the beginning
    - Allows the organization to remain competitive
    - Offers differentiation from competitors

Finding the Right People

- Most challenging part when undertaking a Big Data project
- New Vantage Survey:
  - 95% of senior managers cite people and processes as the main challenges for Big Data projects
  - Business managers need to learn skills to solve business problems with Big Data Analytics
  - The new generation of data analysts must possess a mix of technical and business skills

Finding the Right People (cont.)

- Two alternatives to create an excellent team of data stewards:
  - Organization may hire experts with a computer science or engineering background and train them with business skills
  - Organization may take business undergraduate or MBA professionals and train them with Big Data Analytics tools
- Big Data projects:
  - Provide business insights for better decision-making
  - Offer better customer experience
  - Produce higher quality products or offer better services

Summary and Takeaways

- Big Data is a product of the fourth industrial revolution (4IR)
- Big Data projects are now becoming an established part of business organizations
- The term Big Data was made popular after 2010, when the amount of data existing in the world reached 1 ZB
- In 2030, that number will grow to be 1,000 ZB
- Big Data has three distinguishing characteristics: volume, velocity, and variety
- Big Data Analytics process structured, semistructured, and unstructured datasets
- There are two major pillars in data analysis:
  - Data storage in the form of distributed file systems
  - Data processing in the form of MapReduce and Spark

Summary and Takeaways (cont.)

- Applications of Big Data in many industries
- Importance of Big Data Analytics has increased over time, especially in business organizations
- Two-thirds of companies surveyed view Big Data as critical or very important to their business goals

- Big Data projects present several challenges:
  - Ensuring top management support
  - Deciding between insourcing and outsourcing
  - Creating the right team of data stewards with a mix of technical and business skills

Summary and Takeaways (cont.)

- Big Data projects become feasible when data processing and analytics platforms cannot process the amount, variety, and speed of data generated by the organization
- Big Data should be a part of the organizational strategy and goals
- The ratio of benefits/costs is an approach for prioritizing the importance of Big Data projects
- Other methods include the project scoring model, payback period, net present value, discounted cash flow, internal rate of return, and opportunity cost